# Data Science 500 Assignment

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| QUESTION | EXAMINER MARKS | MODERATOR MARKS | REMARKS |
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## ASSIGNMENT INSTRUCTIONS

*Please tick each box to confirm completion.*

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

**For essay-style assignments, please include the following sections:**

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

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I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

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# 1. Introduction

Book prices and ratings are useful signals for understanding an online catalogue. This project applies a practical data science workflow to product listings on Books to Scrape, a website intended for scraping practice. Python was used to collect book information, prepare a structured dataset, explore patterns through visualizations, and develop a linear regression model to estimate prices.

The analysis focuses on book title, price in pounds sterling (GBP), star rating, stock status and the displayed stock quantity. The aim is to demonstrate an end-to-end workflow and assess whether rating and stock quantity are useful predictors of listed price.

## 2. Project Objectives

- Use requests and BeautifulSoup to collect book listing information from Books to Scrape.
- Create a structured dataset containing at least 200 books.
- Clean and validate the collected fields before analysis.
- Use charts and summary statistics to investigate price and rating patterns.
- Train and evaluate a linear regression model for book-price prediction.
- Communicate findings, limitations and practical recommendations in a professional report.

# 3. Methodology

## 3.1 Data Acquisition

The dataset was collected from https://books.toscrape.com/ using Python's requests library and BeautifulSoup. The scraping code visited the first ten catalogue pages and extracted the title, price, star-rating word, numeric rating, stock status, displayed stock quantity where available, and product URL. Product detail pages were visited to read the stock availability text. The resulting dataset contained 200 records before any subsequent cleaning, based on 20 listings per page across ten pages.

The site is a practice environment for web scraping. Requests were made sequentially, with a short pause between catalogue pages, to avoid unnecessarily rapid access.

## 3.2 Data Cleaning and Preparation

A working copy of the raw dataset was created for cleaning. Duplicate rows were removed; title and stock-status text was stripped of surrounding whitespace; price, rating and stock quantity were converted to numeric types where possible; and records missing essential values such as title, price, rating or stock status were removed. Records with non-positive prices or ratings outside the 1–5 range were excluded. Missing stock quantities were retained as missing values because the source may not provide a quantity in every case.

The cleaned dataset was exported as books_cleaned.csv. The stock quantity field was later imputed with the median inside a modelling pipeline so that missing predictor values could be handled without manually replacing them in the saved dataset.

## 3.3 Exploratory Data Analysis and Visualization

Three visualizations were created: a bar chart of book counts by star rating, a box plot comparing price distributions across ratings, and a scatter/strip plot showing the relationship between rating and price. A price histogram was also used to inspect the overall distribution of listed prices. These plots provide complementary views of the catalogue: frequency by rating, spread and potential outliers in prices, and whether prices visibly change with rating.

Insert the exported or copied versions of your actual graphs in this section before submission. Under each figure, add a numbered caption and a short interpretation based on the graph shown. Do not state exact rating counts or mean prices unless you copy them from your notebook output.

## 3.4 Predictive Modelling

A linear regression model was trained to predict price_gbp using rating and stock_quantity as predictors. The dataset was split into training and test sets using an 80:20 ratio and random_state=42. A scikit-learn pipeline applied median imputation to missing predictor values and then fitted LinearRegression. Performance was evaluated on the test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination ($R^2$). An actual-versus-predicted scatter plot was produced to visually assess prediction errors.

# 4. Results and Discussion

## 4.1 Dataset and visual analysis

The scraping process collected 200 book listings from ten catalogue pages before cleaning. The final number of usable records should be reported from the cleaned dataset's shape in the notebook. The charts should be discussed using the actual counts

and summary statistics printed by the analysis code. The visualizations are intended to identify rating-frequency differences, price variability within each rating group, and whether rating has an obvious relationship with price.

## 4.2 Linear regression results

| Metric | Observed result |
|---|---|
| Training records | 160 |
| Testing records | 40 |
| Mean Absolute Error (MAE) | £13.37 |
| Root Mean Squared Error (RMSE) | £15.20 |
| R-squared ($R^2$) | -0.002 |
| Model intercept | 25.433 |
| Rating coefficient | 0.207 |
| Stock quantity coefficient | 0.522 |

The MAE of £13.37 means that the model's predictions differed from actual test prices by an average absolute amount of about £13.37. The RMSE of £15.20 indicates prediction errors of a similar but somewhat larger magnitude because RMSE penalises larger errors more strongly. The $R^2$ value of -0.002 indicates that the model performed slightly worse on the test set than a baseline that predicts the mean test price.

The actual-versus-predicted plot supports this conclusion: predicted values are clustered around approximately £34–£36 while actual prices span a much wider range. Many points lie far from the diagonal reference line, indicating that the model does not capture much of the variation in book prices.

The fitted equation is approximately: Predicted Price = 25.433 + 0.207 × Rating + 0.522 × Stock Quantity. In this fitted model, the coefficients describe conditional associations, not causal effects. The stock-quantity coefficient should be interpreted cautiously because quantity may be missing for some products and those missing values were imputed during modelling.

## 5. Limitations

Only the first ten catalogue pages were scraped, rather than the complete catalogue.
The model uses only rating and stock quantity; important price-related features such as genre/category, title characteristics, or other product attributes were not included.
Missing stock quantities were imputed using the median, which can reduce the information available from this variable.
The negative $R^2$ indicates poor test-set predictive performance, so the model should not be used for real pricing decisions without further development.
The scraped data reflect the practice website at the time of collection and may change if the site is updated.

## 6. Recommendations

Expand the scraper to collect the complete catalogue and include genre/category information where possible.
Compare a model using rating alone, stock quantity alone, and additional meaningful features; use cross-validation to assess stability.
Check model assumptions and examine residuals, outliers, and prediction errors before relying on predictions.
Keep the cleaned dataset, notebook, and report together so that the analysis is reproducible.
For retail use, treat the current model as a demonstration only because its test performance is poor.

## 7. Conclusion

This project demonstrated an end-to-end data science workflow using Books to Scrape: web data acquisition with BeautifulSoup, data cleaning with pandas, visual exploration, and linear regression using scikit-learn. The model achieved an MAE of £13.37, RMSE of £15.20 and $R^2$ of -0.002 on a 40-record test set. These results show that rating and stock quantity alone did not provide useful price predictions in this experiment. A broader dataset and additional product features are recommended for future work.

Before submission, insert the actual graphs, add any verified summary statistics from the notebook, and ensure that the cleaned row count stated in the report matches the final CSV.

# References


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